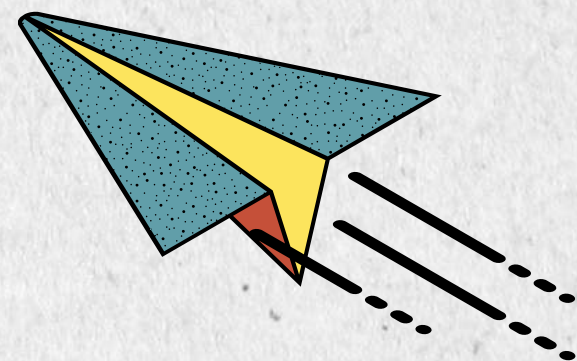




HPDP PROJECT 2 SECTION 02

SHOPEE APPLICATION REAL-TIME SENTIMENT ANALYSIS USING APACHE SPARK AND KAFKA

GROUP BigPotato



PROBLEM



Massive volume of Google Play reviews for Shopee

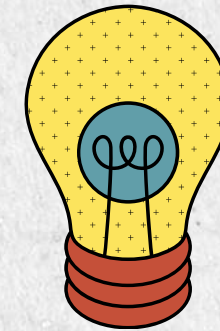


Manual analysis is inefficient

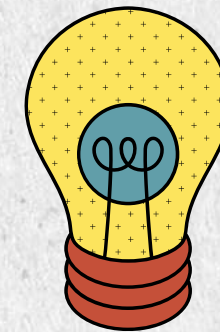


Need timely customer insights

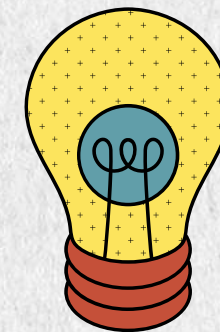
SOLUTION



NLP for text preprocessing

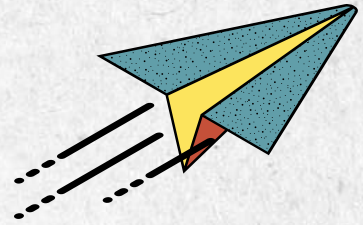


Machine learning for sentiment classification

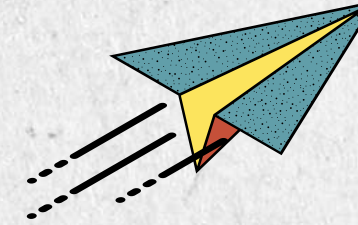


Real-time analytics using Kafka & Spark

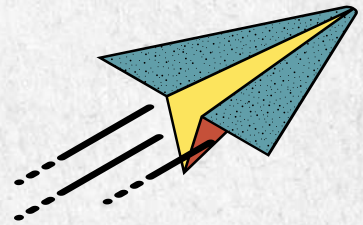
PROJECT OBJECTIVES



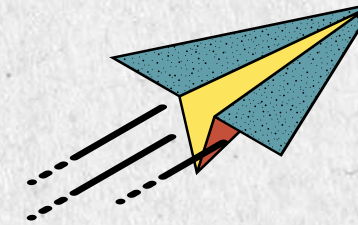
Collect Shopee Malaysia Google Play reviews



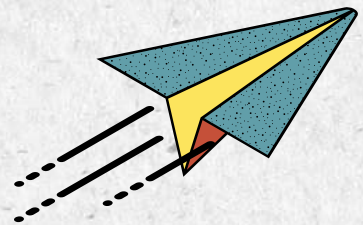
Build Kafka–Spark streaming pipeline



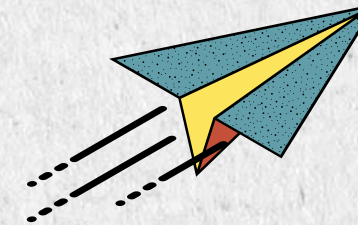
Perform NLP preprocessing



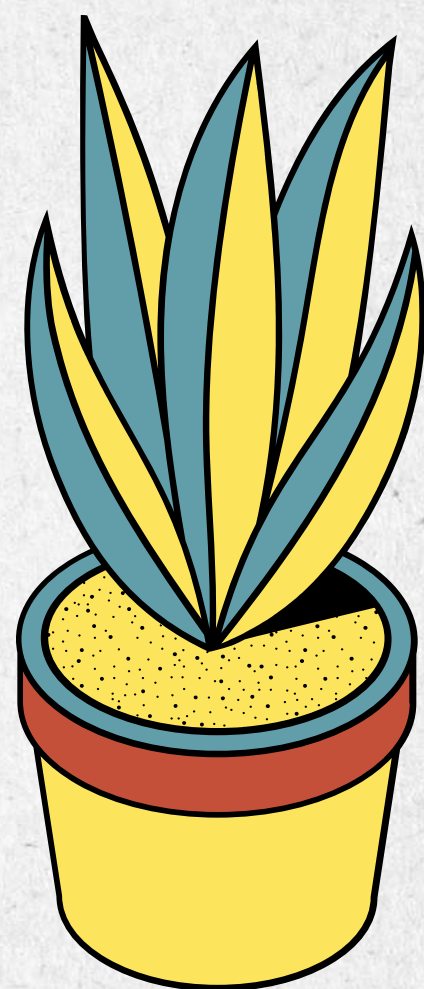
Visualize results using Elasticsearch result



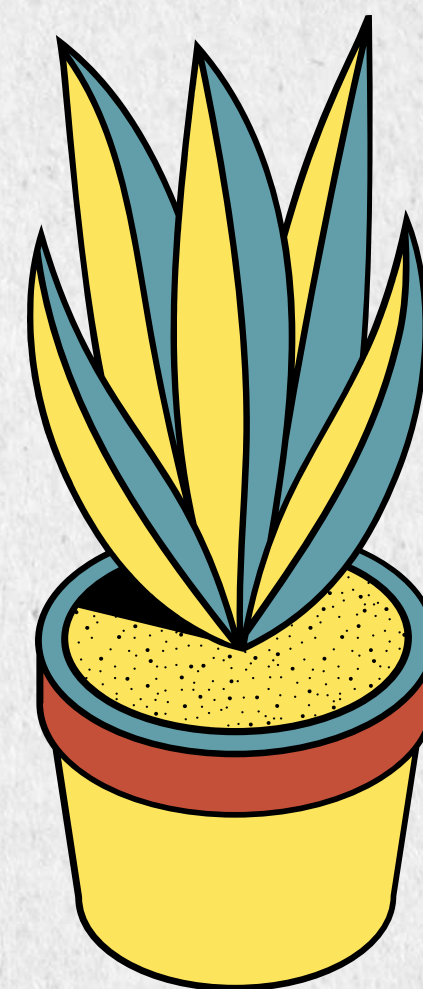
Train & compare sentiment models



Compare Batch vs Streaming performance



DATA COLLECTION

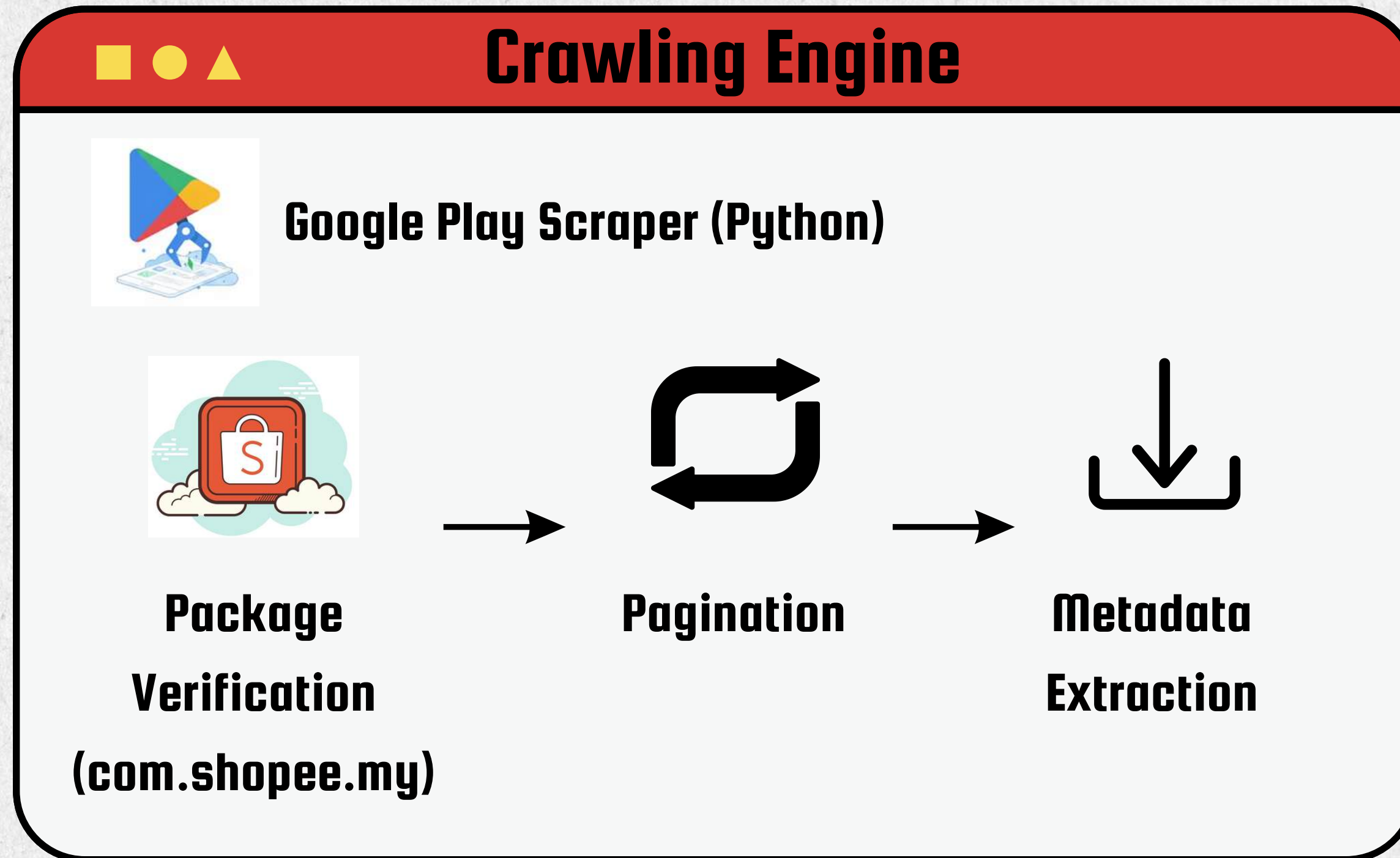


COLLECTION PROCESS

Data Source



- Public customer reviews and ratings
- Real-world Malaysian customer feedback



Output

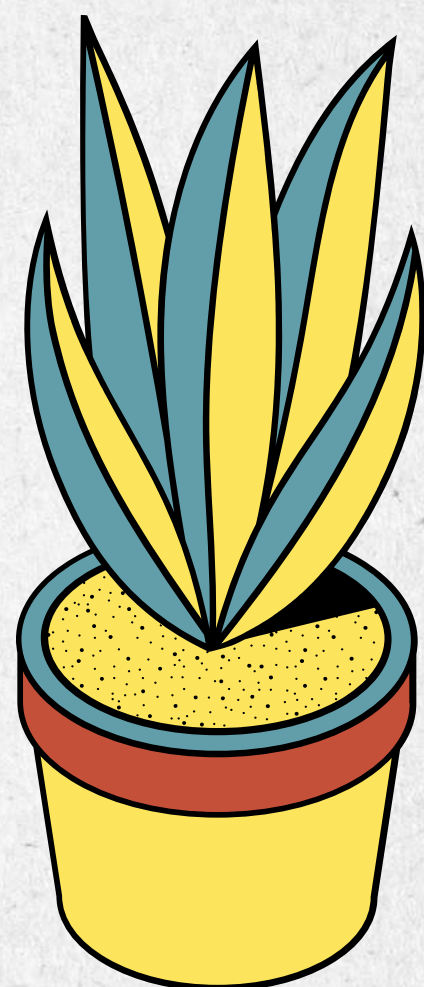


- 10000 rows records
- 6 attributes
- CSV file

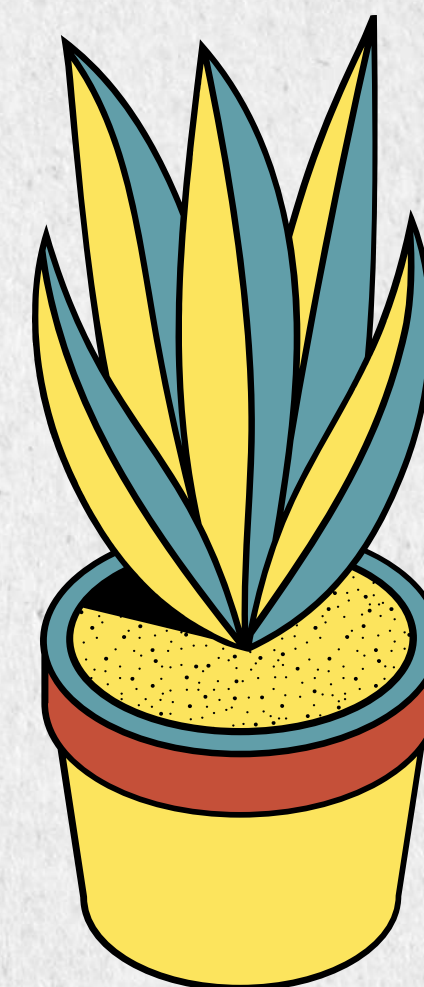
RAW DATA



No	Attribute	Description	Example
1	Review ID	Unique review identifier	12c34620-7baa-4041-85a9-2fc1ff43a864
2	Rating	1–5 stars	5
3	Review Date	Submission date	6/28/2026
4	App Version	App version	3.56.15
5	Thumbs Up	Helpful votes	5
6	Original Text	Customer review	"Fast delivery and many vouchers."

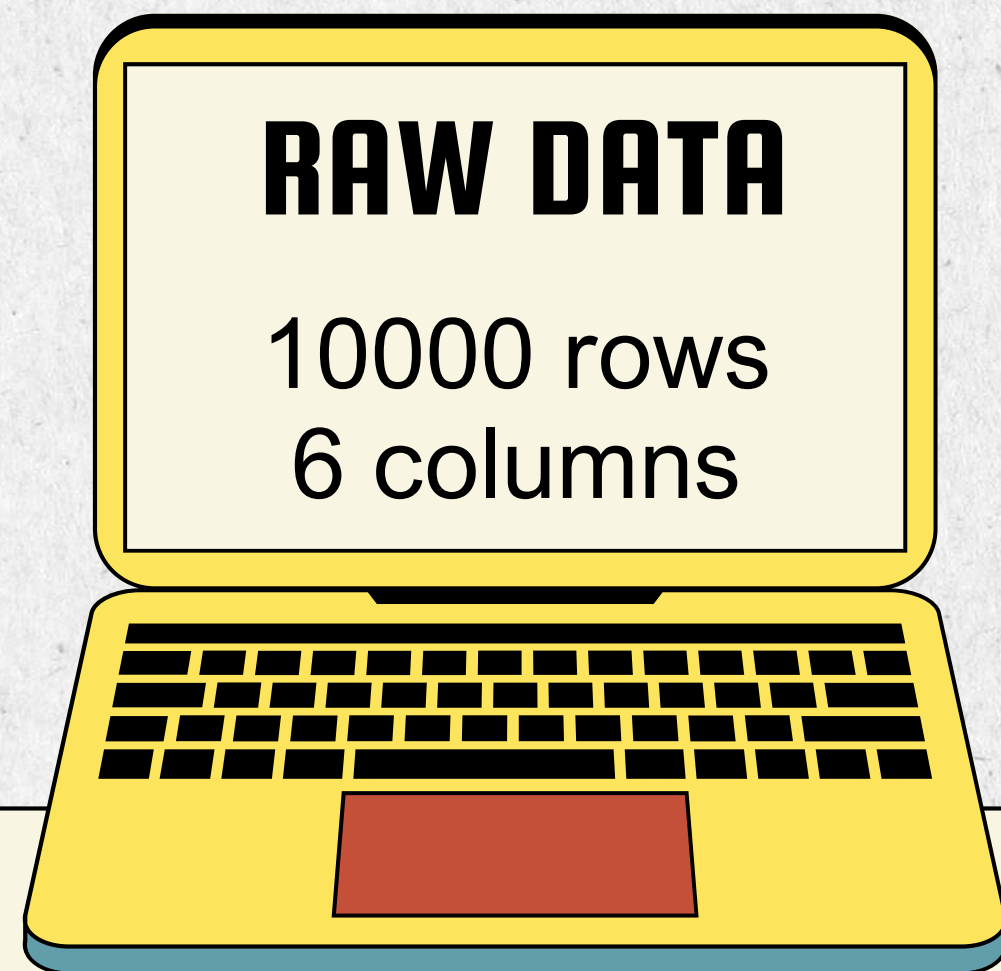


DATA PROCESSING



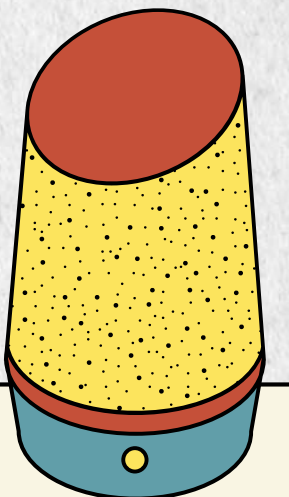
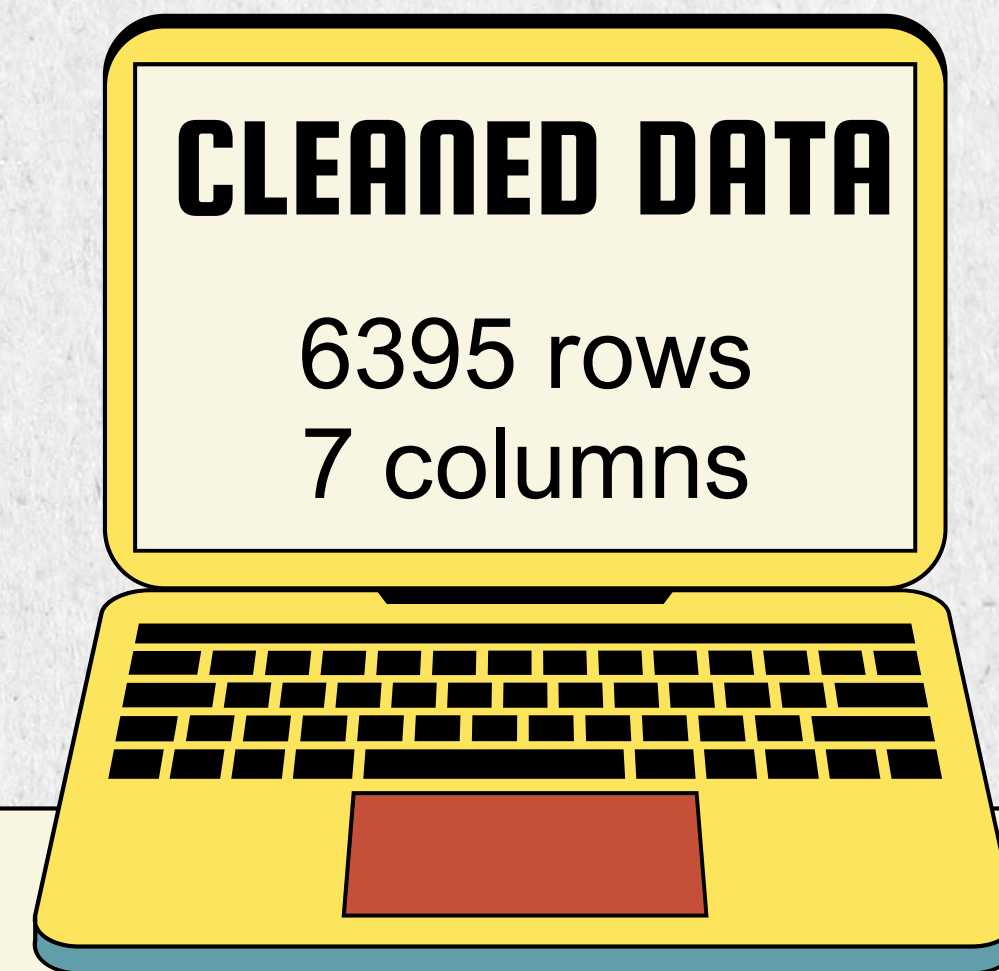
DATA CLEANING

- Removed duplicate review IDs and duplicate review texts
- Removed missing, empty, and low-information reviews (less than 3 words)
- Reduced dataset from 10,000 to 6,395 cleaned reviews



NLP PREPROCESSING

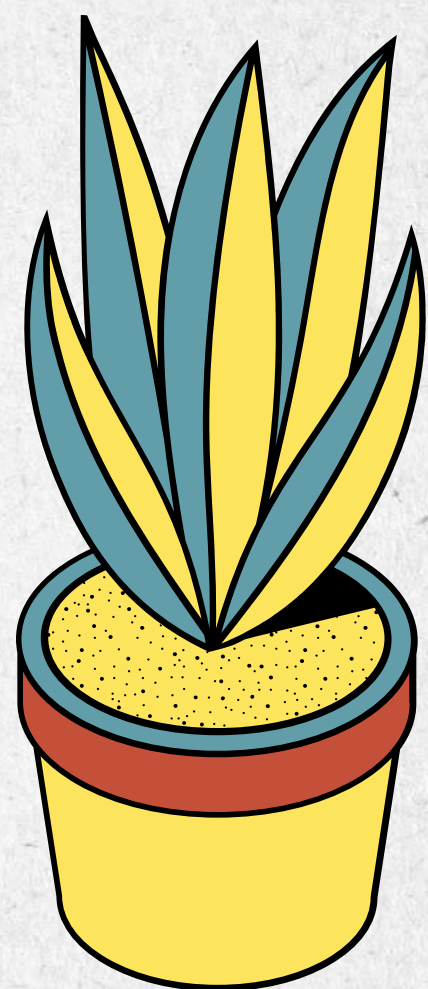
- Converted all text to lowercase
- Text cleaning (URL, punctuation, emoji, numbers)
- Tokenized review text into individual words
- Removed English and Malay stop words
- Applied English lemmatization
- Generated a new cleaned_text column while preserving the original review



CLEANED DATA

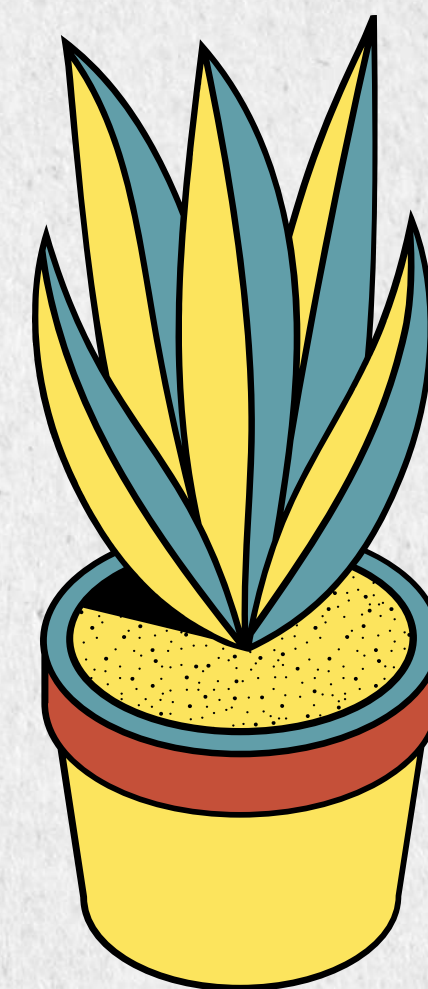


No	Attribute	Description	Example
1	Review ID	Unique review identifier	12c34620-7baa-4041-85a9-2fc1ff43a864
2	Rating	1–5 stars	5
3	Review Date	Submission date	6/28/2026
4	App Version	App version	3.56.15
5	Thumbs Up	Helpful votes	5
6	Original Text	Customer review	shopee very nice and very easy
7	Cleaned Text	Review after NLP preprocessing	shopee nice easy



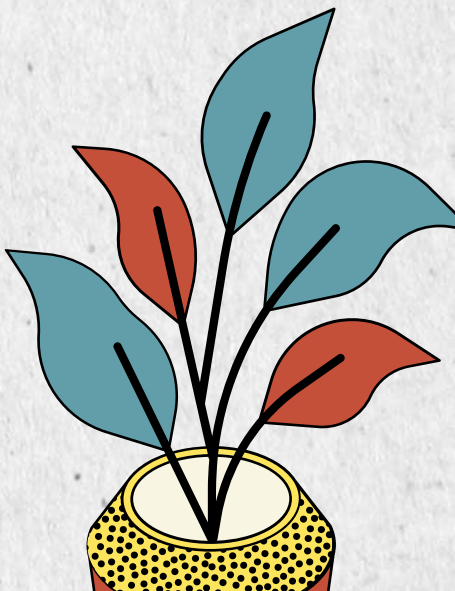
SENTIMENT MODEL

DEVELOPMENT



MODEL DEVELOPMENT

- **Naive Bayes** — TF-IDF + Laplace smoothing ($\alpha=0.1$)
- **Logistic Regression** — TF-IDF + balanced class weighting ($C=0.5$)
- **Linear SVM** — TF-IDF + balanced class weighting ($C=0.5$)

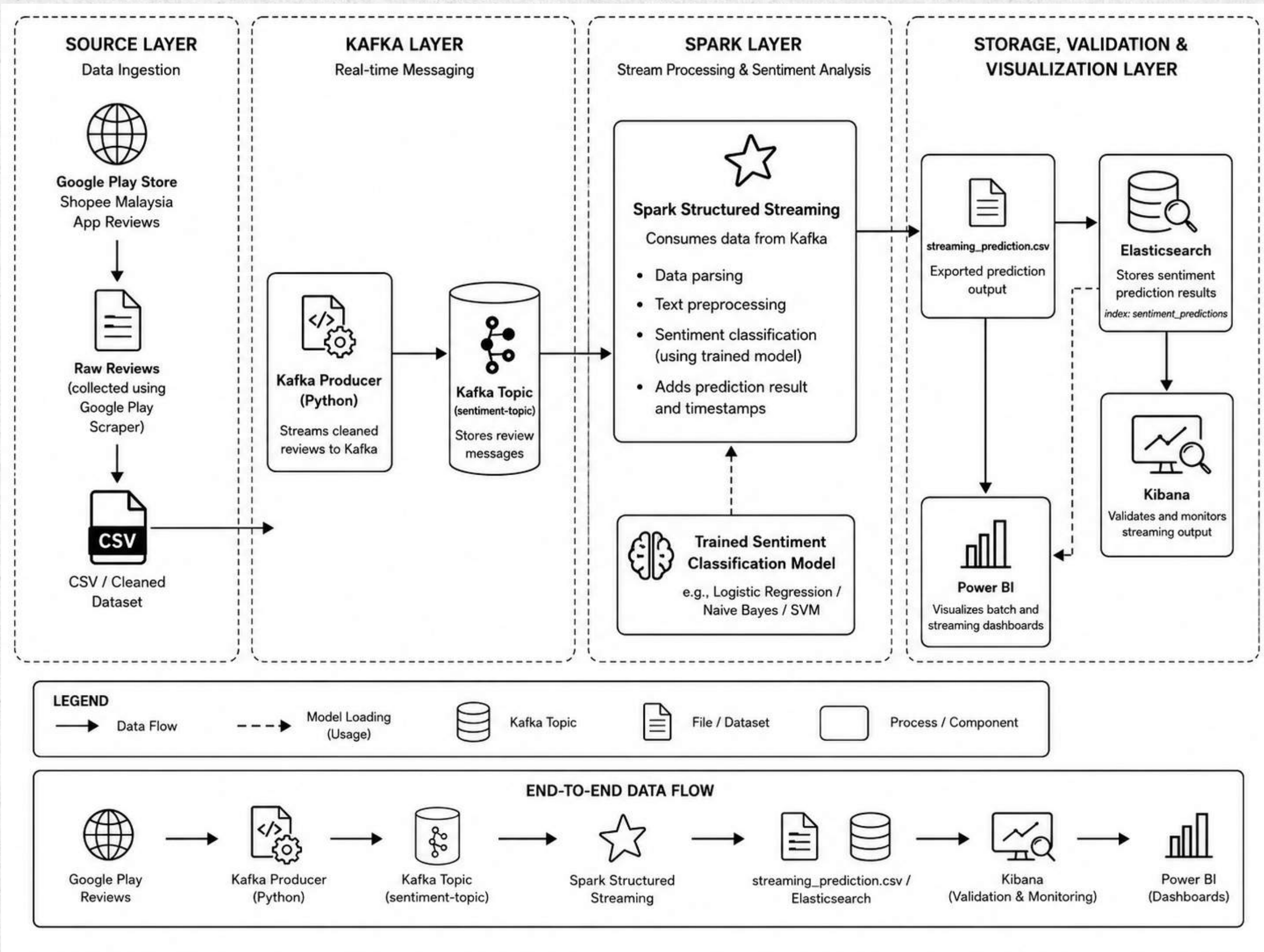


MODEL COMPARISON

Model	Accuracy	Weighted F1	Macro F1
Naive Bayes ★	88.19%	0.8663	0.59
Logistic Regression	84.75%	0.8475	0.63
Linear SVM	86.40%	0.8592	0.64

 **Best Model Selected: Multinomial Naive Bayes**
Reason: Achieved the highest Weighted F1-Score (0.8663) and Accuracy (88.19%) while ensuring high computational efficiency for the real-time Spark streaming pipeline.

APACHE SYSTEM ARCHITECTURE



The real-time pipeline connects Kafka, Spark Structured Streaming, Elasticsearch, Kibana, and Power BI to process Shopee review sentiment continuously.

REAL-TIME PIPELINE IMPLEMENTATION

1. Kafka Data Ingestion

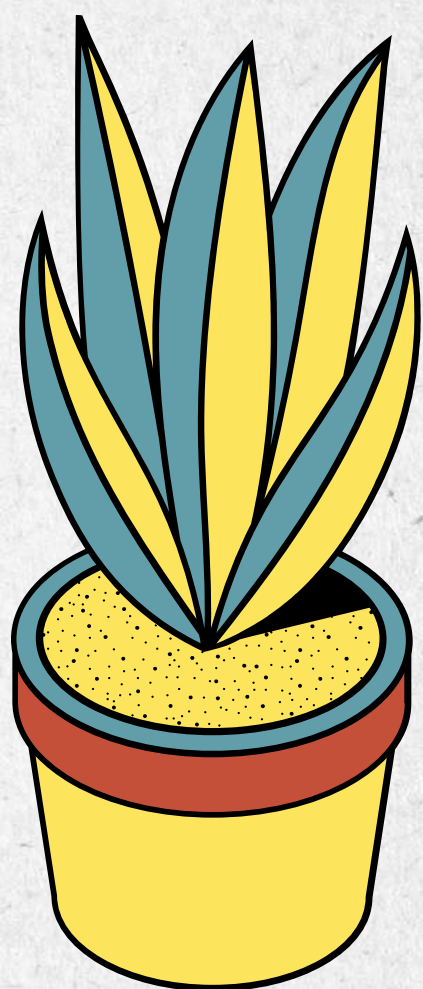
- Reads cleaned review data row by row
- Converts each review into JSON message
- Sends messages to sentiment-topic

2. Spark Structured Streaming Processing

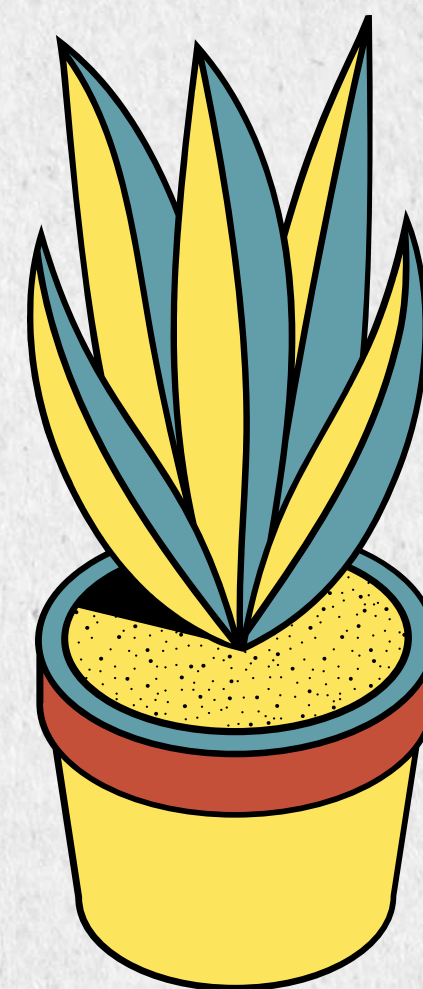
- Consumes messages from Kafka
- Applies trained sentiment model to cleaned_text
- Processes data in micro-batches

3. Output and Storage

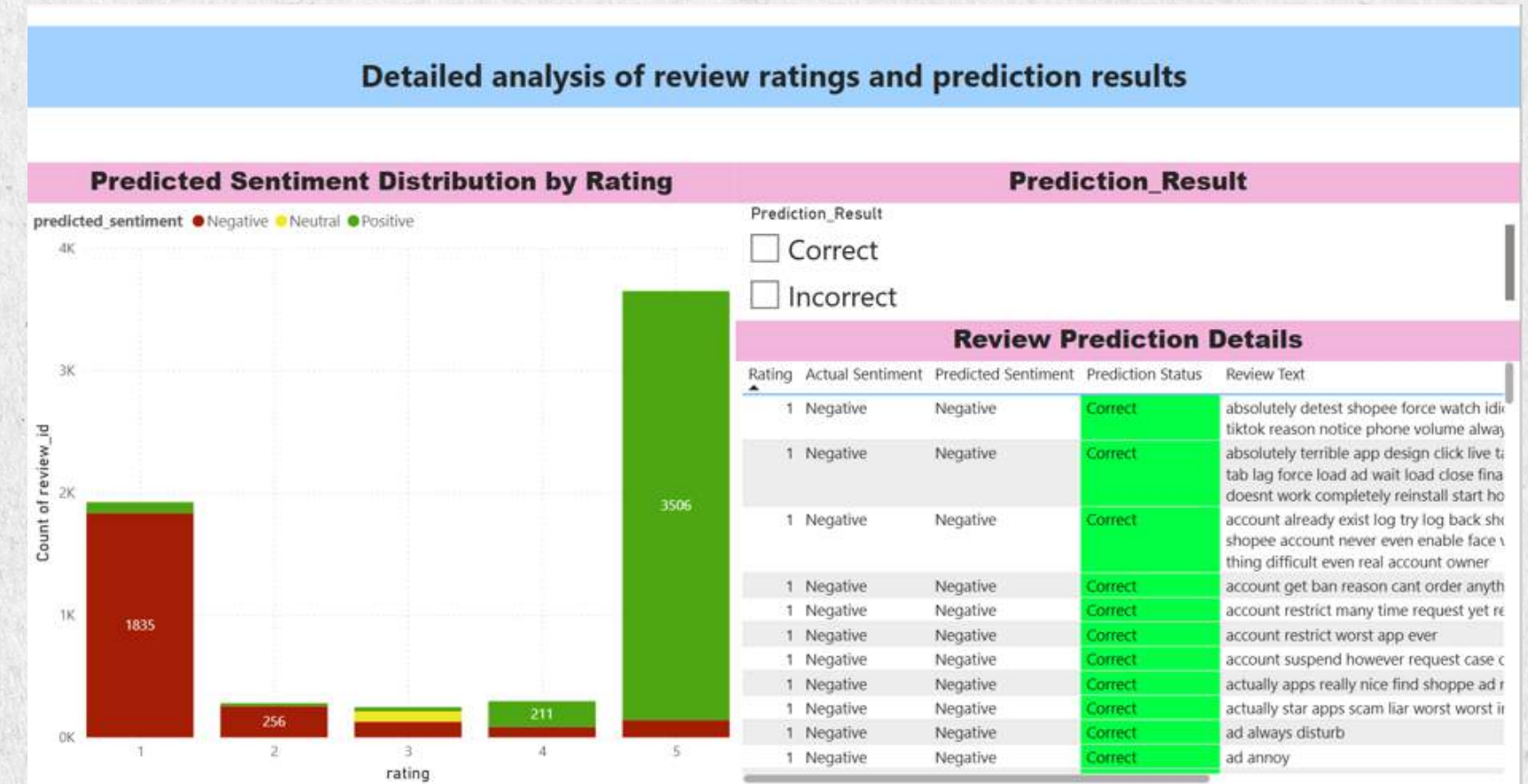
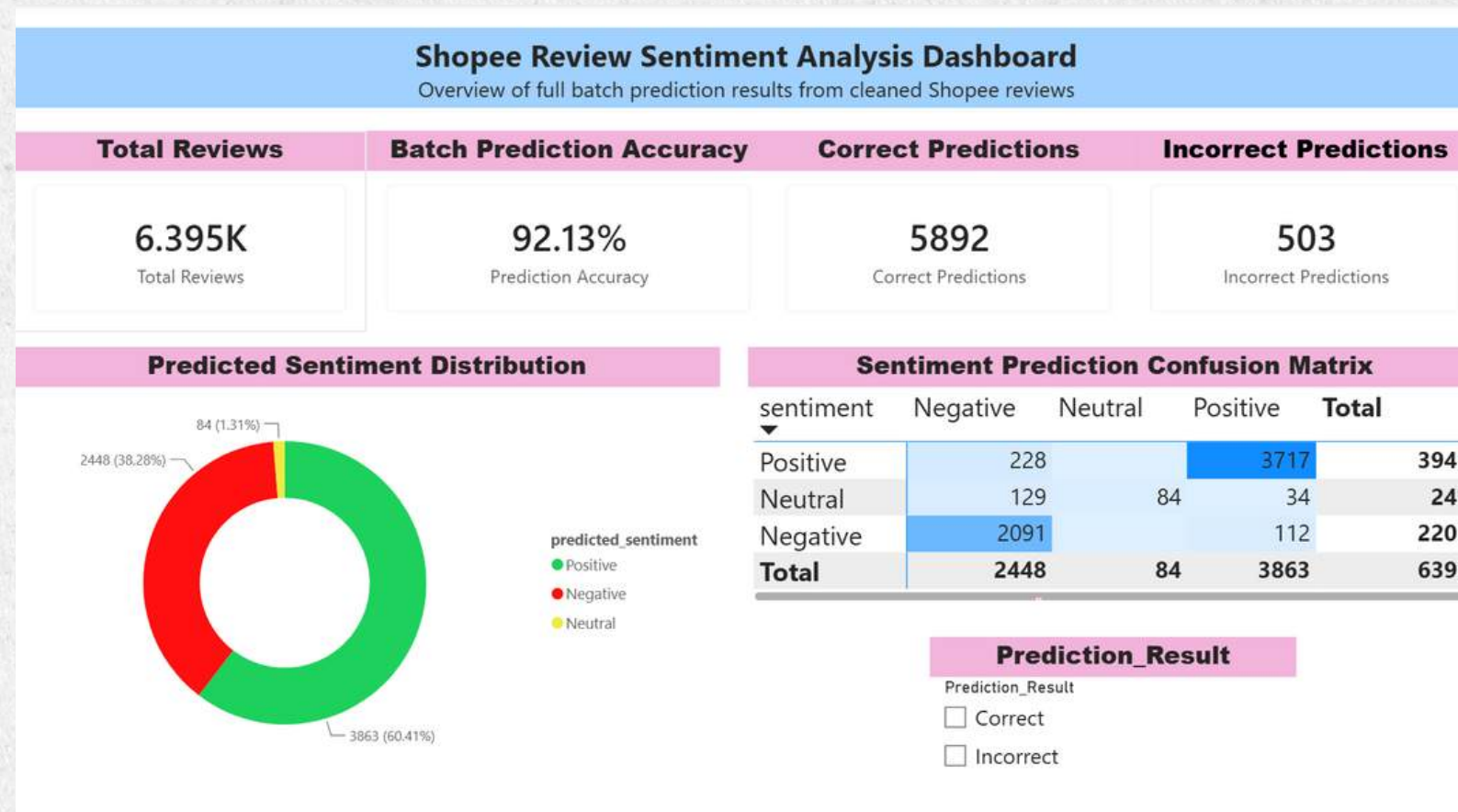
- Saves results to streaming_prediction.csv
- Stores results in Elasticsearch
- Supports Kibana validation and Power BI dashboard



DASHBOARD

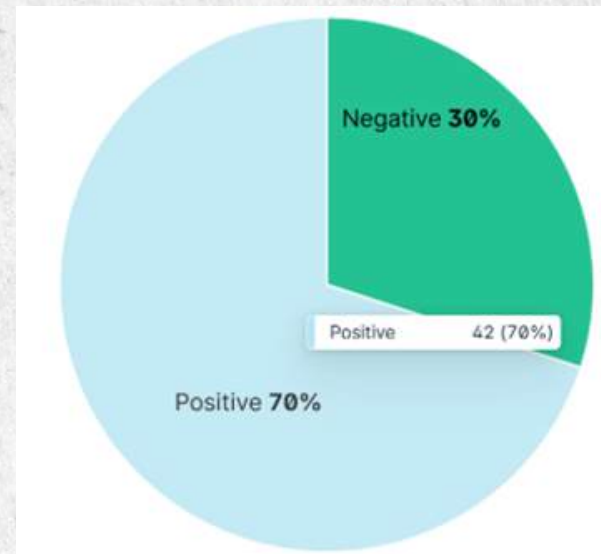


POWER BI DASHBOARD

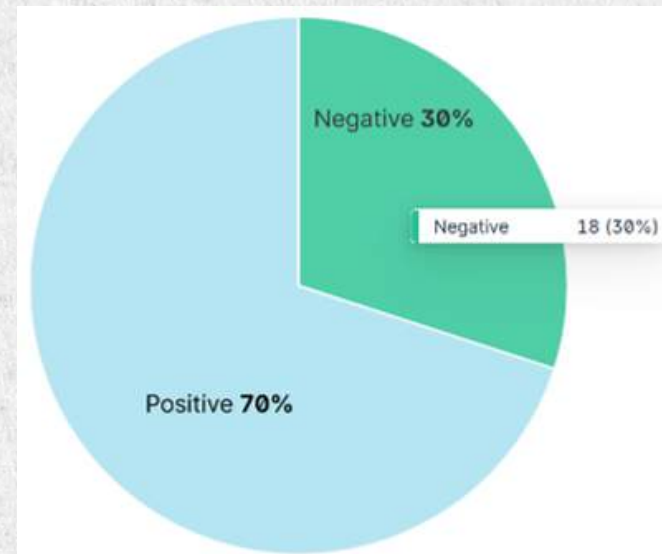


- 6395 reviews were analysed with 92.13% batch accuracy.
- Positive sentiment was dominant while neutral sentiment was the hardest to classify.
- Detailed review filtering helps identify correct and incorrect predictions.
- Both correct and incorrect details could be analyse at the second dashboard
- Rating 1,2 mostly are Negative reviews while rating 4,5 mostly are Positive review

KIBANA DASHBOARD



Streaming Sentiment Distribution in Kibana



- 60 streaming messages were used and visualized in Kibana.
- Streaming sentiment consisted of 70% positive and 30% negative predictions.
- Due to small streaming samples, only positive and negative samples are selected.
- Spark processed the messages in multiple micro-batches over time.

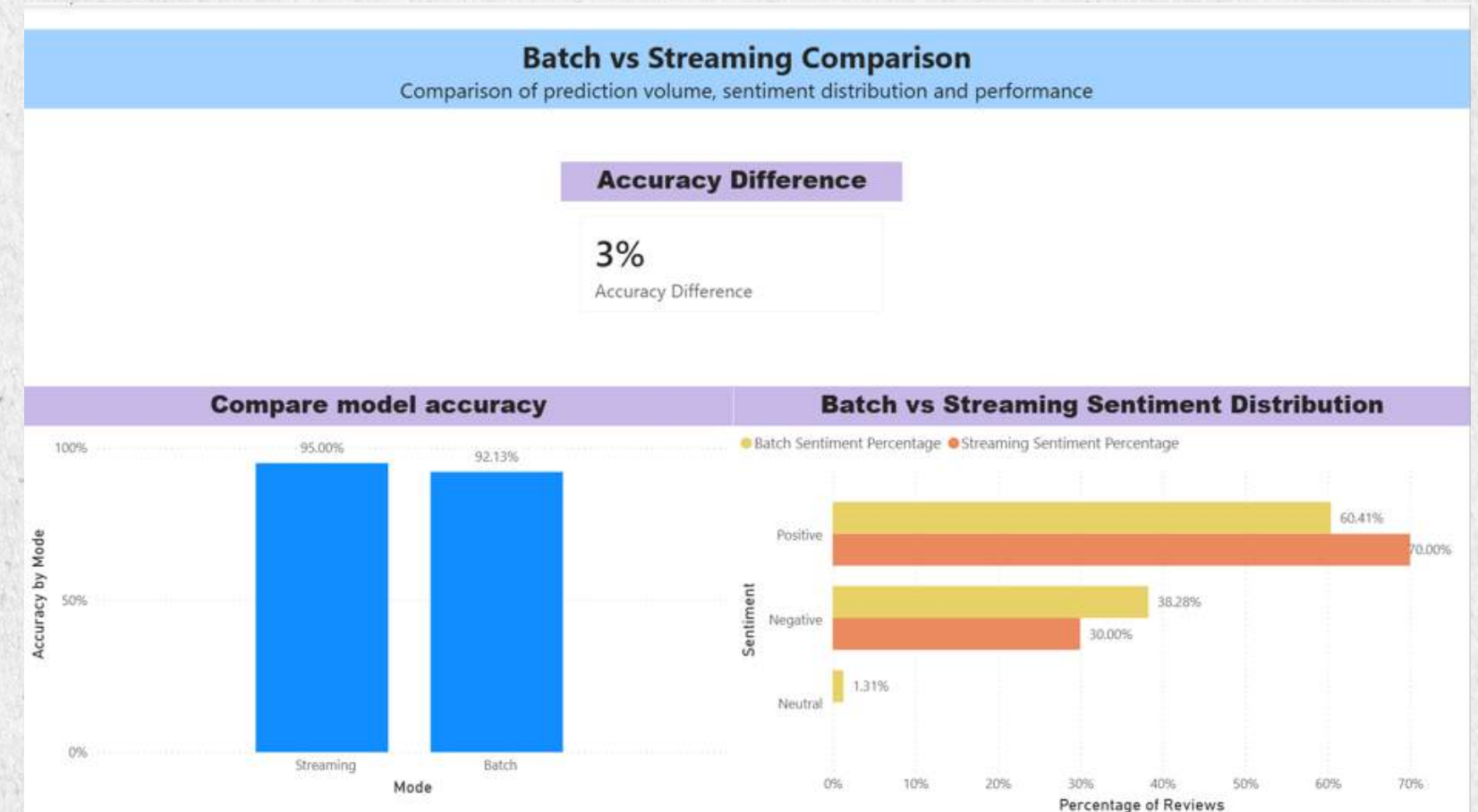
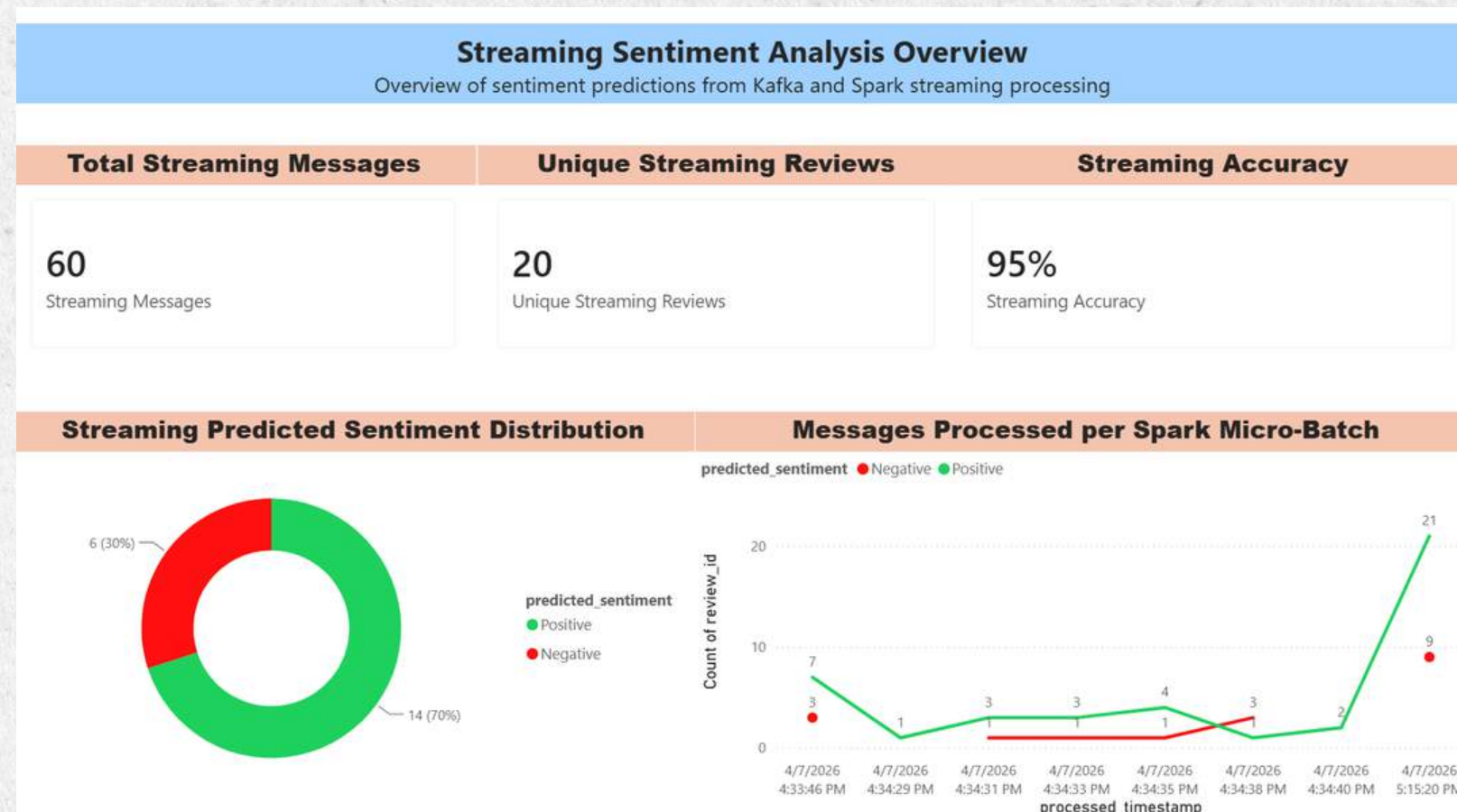


Predicted Sentiment by Review Rating

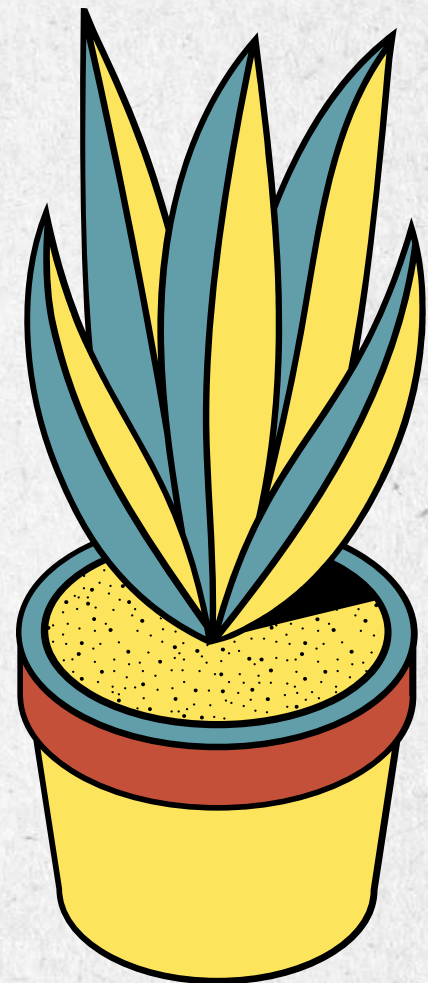


Sample Reviews Processed by Timestamp

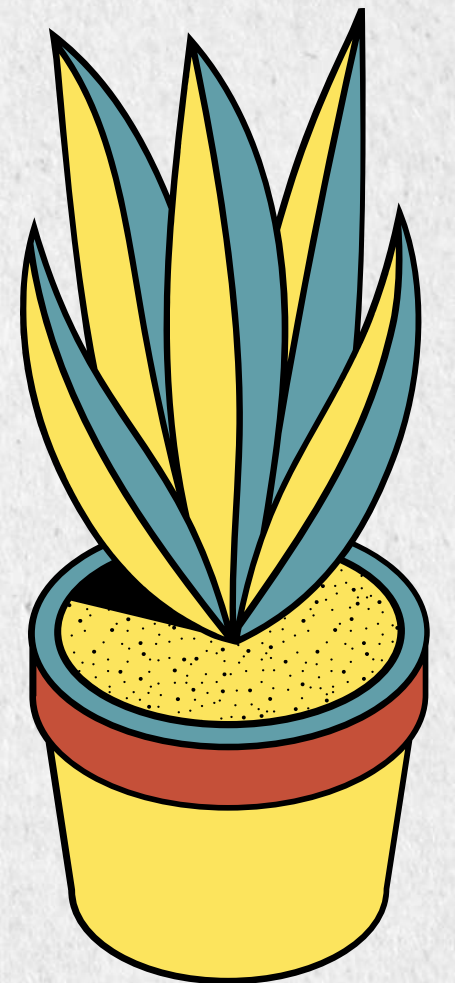
POWER BI DASHBOARD



- The streaming dashboard proves that the Kafka and Spark pipeline can process and visualize review sentiment in near real time.
- Neutral sentiment does not appeared in streaming processing
- 3% of accuracy difference
- Positive sentiment was dominant in both batch and streaming results.
- Streaming showed a higher positive sentiment percentage at 70%, compared with 60.41% in batch processing.



OPTIMISATION & COMPARISON



BATCH AND STREAMING PERFORMANCE COMPARISON

Performance Metric	Batch Processing	Streaming Processing
Total records processed	6395 records	6395 records
Total processing time	0.3968 seconds	9.0977 seconds
System throughput	16,116.71 records/second	702.92 records/second
CPU utilization rate	95.90%	6.01%
Peak memory consumption	236.23 MB	240.29 MB
Classification accuracy	88.19%	95.00% based on 20 streaming sample

OPTIMIZATION AND COMPARISON

Model Optimization

- Neutral sentiment was difficult to classify
- Collect more neutral reviews to balance the dataset
- Apply resampling or class-balancing techniques
- Try stronger models such as Logistic Regression, Linear SVM, BERT, or DistilBERT

Architecture Optimization

- Use larger streaming sample for fairer testing
- Increase Kafka partitions for better parallel processing
- Tune Spark micro-batch interval and memory settings
- Add checkpointing, error handling, and monitoring

Overall Discussion

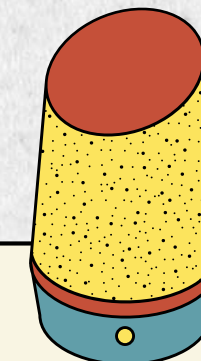
- Batch processing was faster for fixed dataset analysis
- Streaming processing is more suitable for near real-time monitoring
- Future work should improve neutral sentiment classification and pipeline scalability

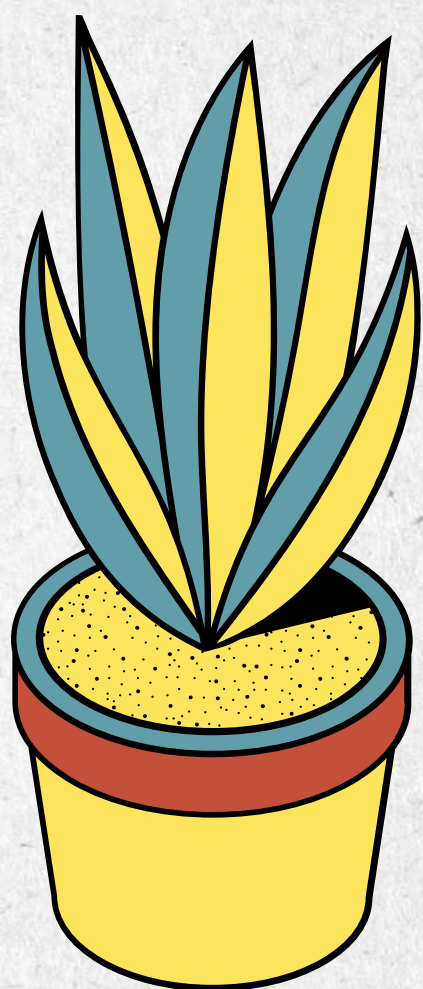
SUMMARY

- Built an end-to-end sentiment pipeline
- Cleaned 10,000 records into 6,395 rows
- Produced a cleaned dataset with 25,209 rows and 7 columns.
- Compared three machine learning approaches: Multinomial Naive Bayes, Logistic Regression and Linear SVM.
- Multinomial Naive Bayes achieved the best performance
- Successful Stream & Batch Processing

FUTURE WORKS

- Benchmark using larger & balanced datasets
- Explore advanced transformer-based models
- Optimise streaming pipelines with real-world data
- Enhance system telemetry & cluster allocations
- Implement production-grade reliability





THANK YOU

